Pet Care Clinic Database

Database Design & MySQL Implementation

Selected academic project documentation

Public portfolio edition

Prepared from the original university team project report.

Pages containing identifying information or contact records were excluded.

Team Responsibilities

- Retaj Ayman Almehmadi: Prepared the ER Diagram and helped identify the relationships.
- Nada Abdulaziz Alsubhi: Worked on the relational schema and database structure.
- Layan Ayman Alamoudi: Wrote and tested the SQL queries and views.
- Revan Qaseem: Inserted sample data and checked the final outputs.

Conceptual and Logical Design

The Pet Care Clinic database was designed to organize information about owners, pets, appointments, and vaccinations.
The main entities are Owners, Pets, Appointments, and Vaccinations.
Each owner can have one or more pets. Each pet can have many appointments and vaccinations.

Business Rules:

Entities
- Owners
- Pets
- Appointments
- Vaccinations

Attributes
Owners
- OwnerID (Primary Key)
- OwnerName
- PhoneNumber
- Address
- Email

Pets
- PetID (Primary Key)
- PetName
- Species
- Breed
- Gender
- OwnerID (Foreign Key)

Appointments
- AppointmentID (Primary Key)
- AppointmentDateTime
- Reason
- PetID (Foreign Key)

Vaccinations
- VaccinationID (Primary Key)
- VaccinationName
- DateGiven
- NextDueDate
- PetID (Foreign Key)

Relationships
- Owners to Pets → One-to-Many (1:M)
- Pets to Appointments → One-to-Many (1:M)
- Pets to Vaccinations → One-to-Many (1:M)

Entities, Attributes, and Relationships

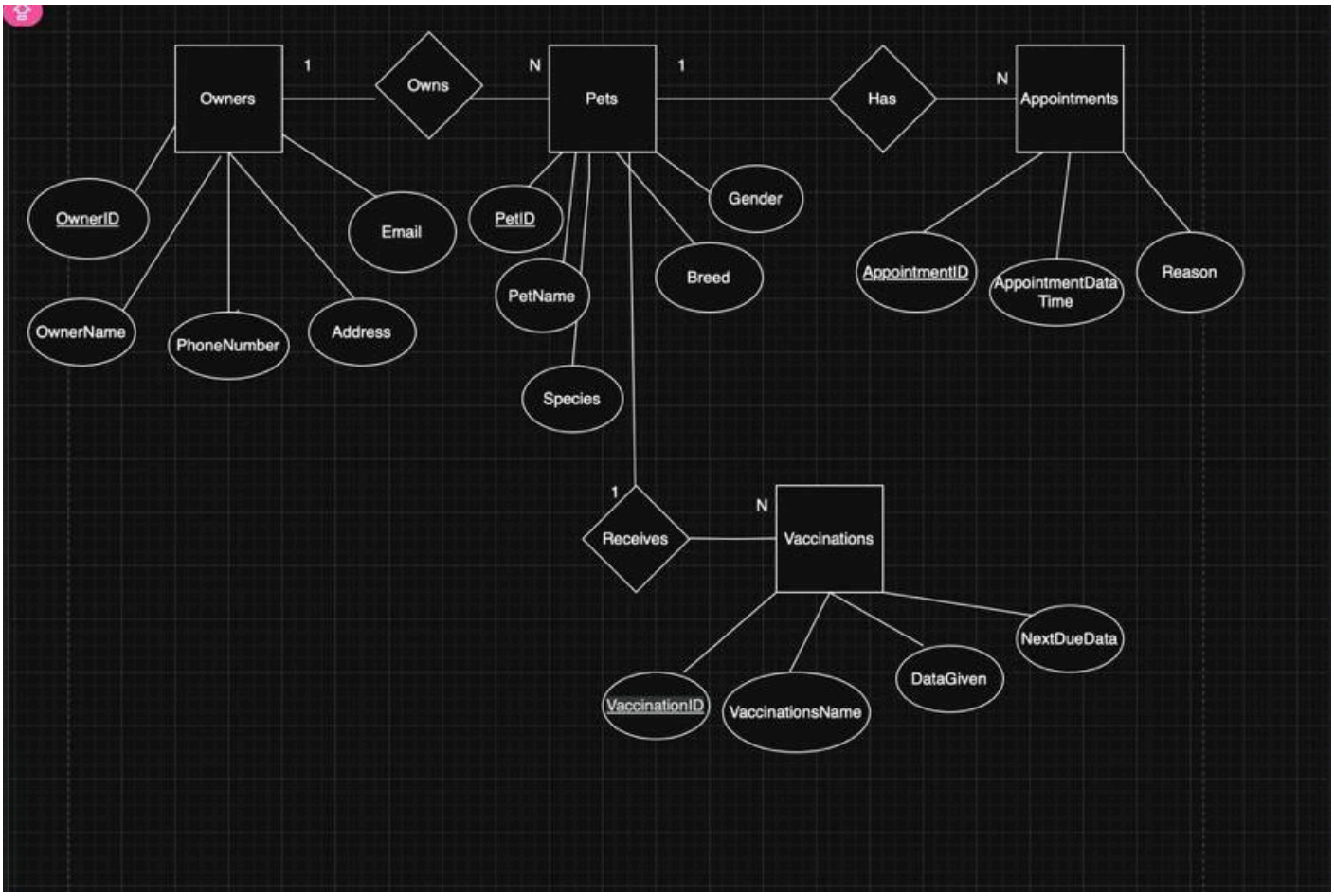
Entities and Attributes

Owners: ownerID, ownerName, phoneNumber, address, email
Pets: petID, petName, species, breed, gender, ownerID
Appointments: appointmentID, appointmentDateTime, reason, petID
Vaccinations: vaccinationID, vaccinationName, dateGiven, nextDueDate, petID

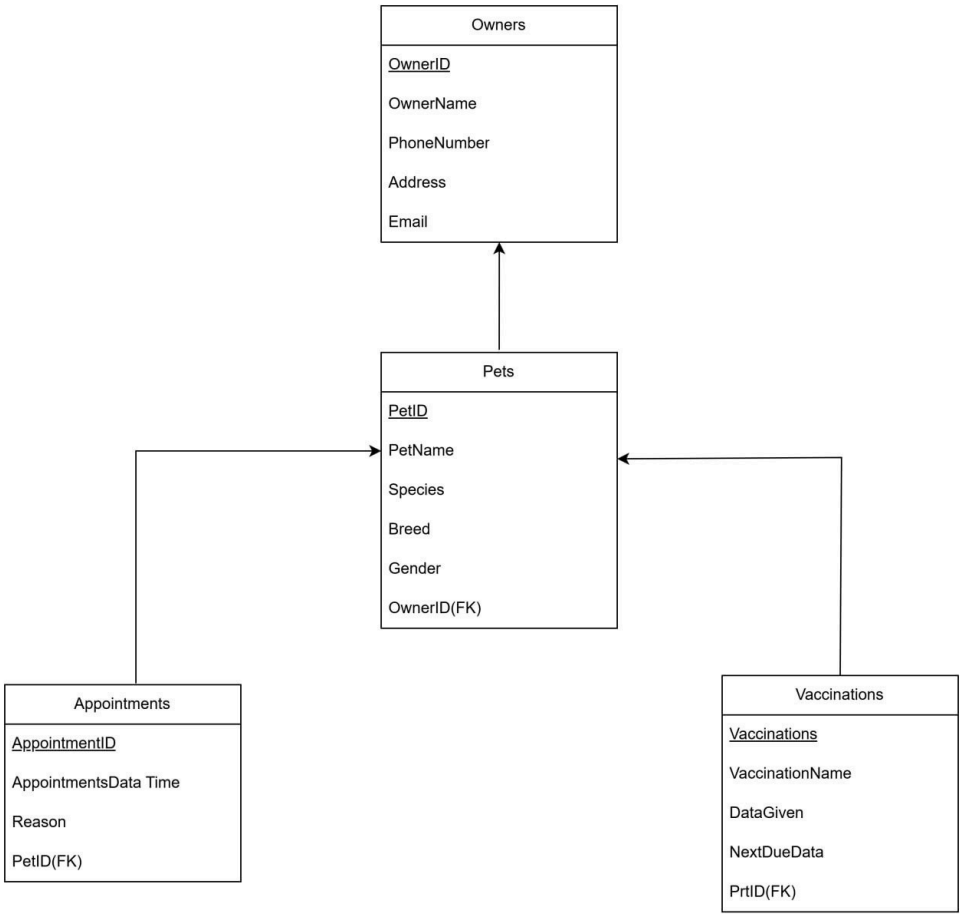
Relationships

Owners to Pets: One-to-Many
Pets to Appointments: One-to-Many
Pets to Vaccinations: One-to-Many

ER Diagram

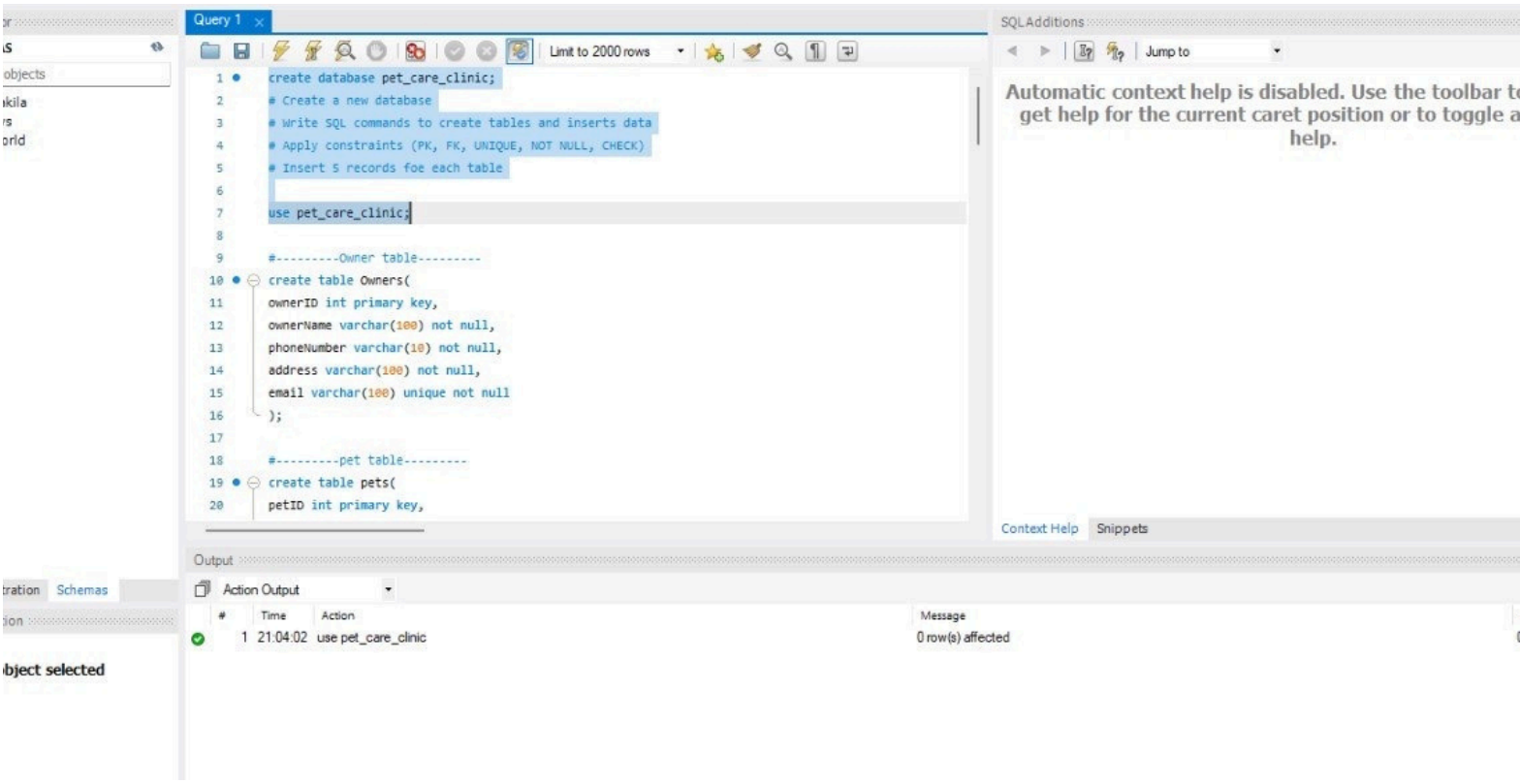

Owners
1
Owns
N
Pets
1
Has
N
Appointments
OwnerID
Email
PetID
Gender
AppointmentID
AppointmentData Time
Reason
OwnerName
PhoneNumber
Address
PetName
Breed
Species
1
Receives
N
Vaccinations
NextDueData
DataGiven
VaccinationID
VaccinationsName

Relational Schema



Database Implementation

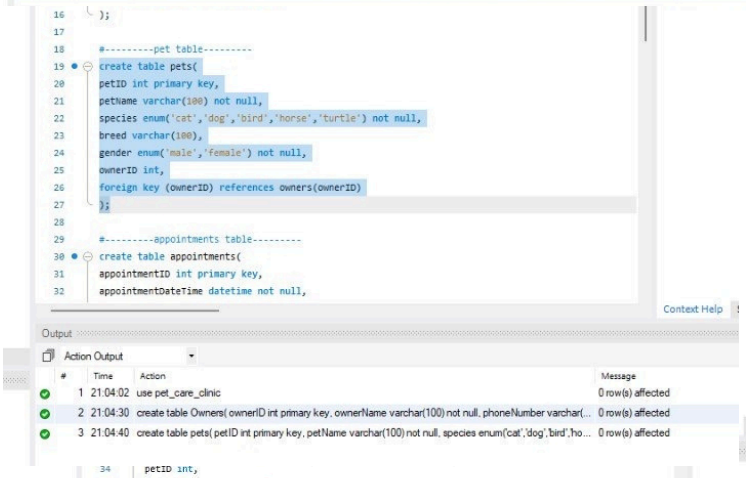
The database was created using MySQL.
The database name is pet_care_clinic.
Four tables were created: Owners, Pets, Appointments, and Vaccinations.



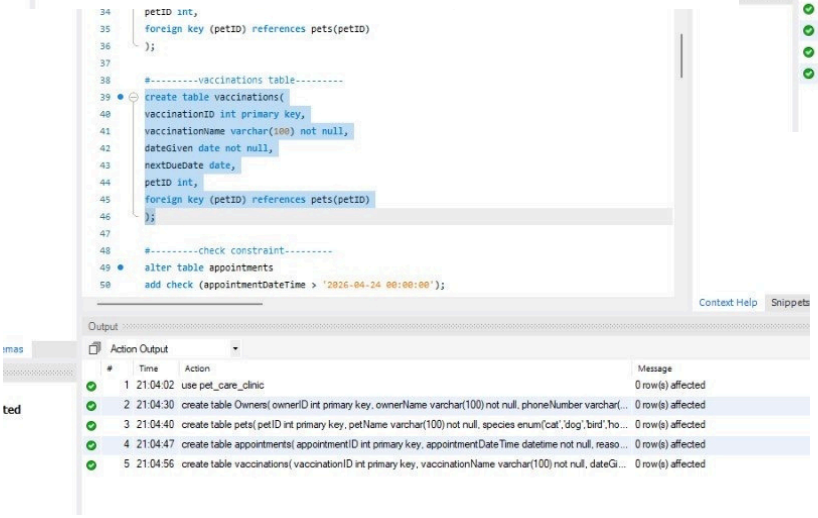
Tables Creation

The tables were created using CREATE TABLE commands.
Each table contains a primary key, and foreign keys were used to connect the tables together.









Data Types and Constraints

The database uses different data types such as INT, VARCHAR, ENUM, DATETIME, and DATE.
The constraints used in this project are PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, and CHECK.

Appointment table:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
appointmentID	int	NO	PRI	NULL	
appointmentDateTime	datetime	NO		NULL	
reason	enum('checkup','illness','vaccination','surgury')	NO		NULL	
petID	int	YES	MUL	NULL	

Owner table:

Field	Type	Null	Key	Default	Extra
ownerID	int	NO	PRI	NULL	
ownerName	varchar(100)	NO		NULL	
phoneNumber	varchar(10)	NO		NULL	
address	varchar(100)	NO		NULL	
email	varchar(100)	NO	UNI	NULL	

Pet table:

Vaccination table:

Result Grid | Filter Rows: | Export: | Wrap C

Field	Type	Null	Key	Default	Extra
vaccinationID	int	NO	PRI	NULL	
vaccinationName	varchar(100)	NO		NULL	
dateGiven	date	NO		NULL	
nextDueDate	date	YES		NULL	
petID	int	YES	MUL	NULL	

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
petID	int	NO	PRI	NULL	
petName	varchar(100)	NO		NULL	
species	enum('cat','dog','bird','horse','turtle')	NO		NULL	
breed	varchar(100)	YES		NULL	
gender	enum('male','female')	NO		NULL	
ownerID	int	YES	MUL	NULL	

Result 13

```
48  #---------check constraint---------
49  alter table appointments
50  add check (appointmentDateTime > '2026-04-24 00:00:00');
51
52  #---------inserting data---------
53
54  # owners data:
55  insert into owners VALUES
56  (1,'khaled alotibi','0557899485','alnmail','khal5112ed@icloud.com'),
57  (2,'noor nasser','0595842184','alnasser','noor6@nasa4email.com'),
58  (3,'mohamed almossab','05555515216','alshuqea','moha23hhm@gmail.com'),
59  (4,'aya imam','0559257817','alnamli','ayaanphysics22@icloud.com'),
60  (5,'amal alzahrani','0556615218','alnuzhah','amalthequeen@gmail.com');
61
62  # pets data:
```

CHECK Constraint

Output

Action Output

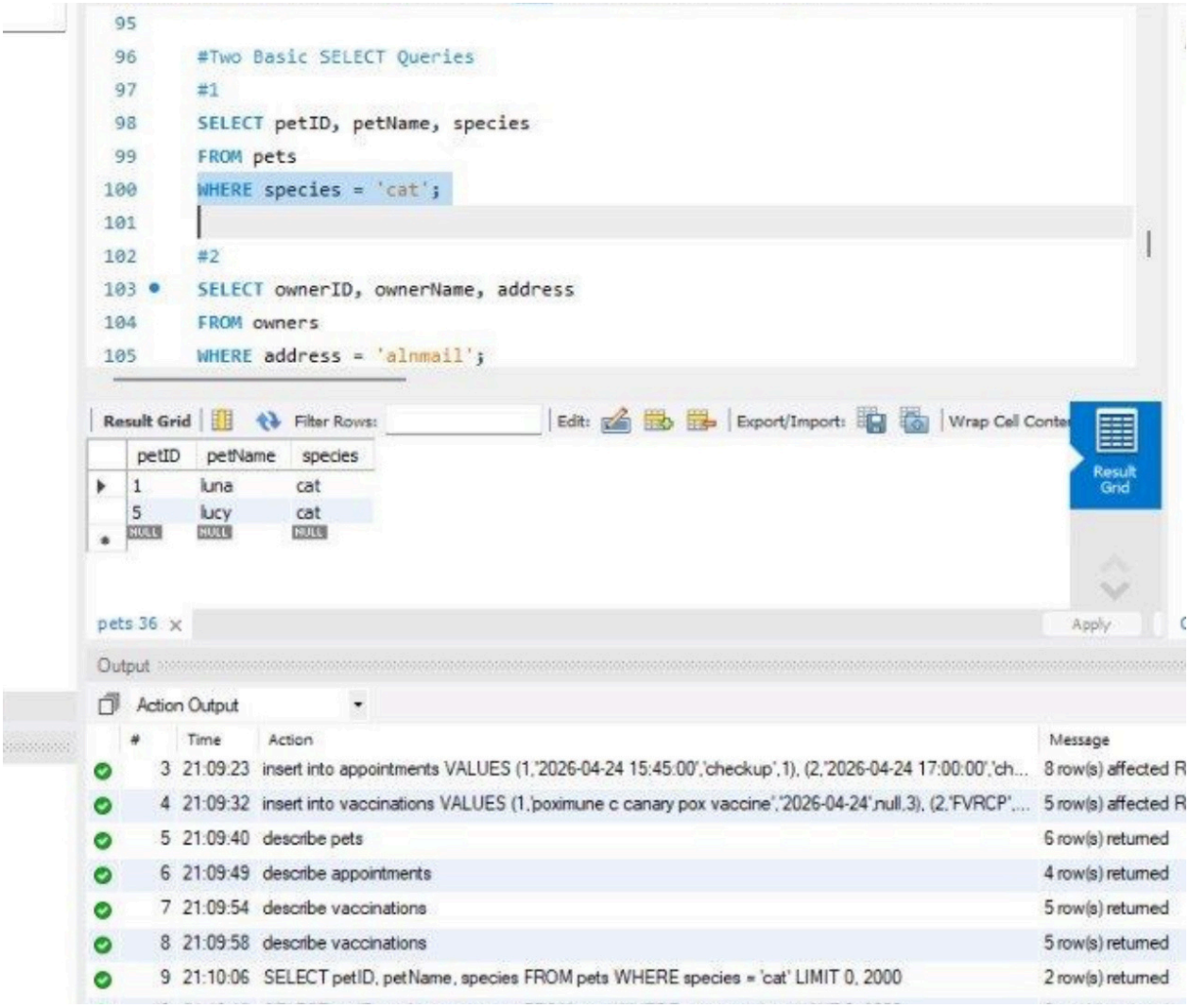
#	Time	Action	Message
1	21:04:02	use pet_care_clinic	0 row(s) affected
2	21:04:30	create table Owners(ownerID int primary key, ownerName varchar(100) not null, phoneNumber varchar(...	0 row(s) affected
3	21:04:40	create table pets(petID int primary key, petName varchar(100) not null, species enum('cat','dog','bird','ho...	0 row(s) affected
4	21:04:47	create table appointments(appointmentID int primary key, appointmentDateTime datetime not null, reaso...	0 row(s) affected
5	21:04:56	create table vaccinations(vaccinationID int primary key, vaccinationName varchar(100) not null, dateGi...	0 row(s) affected
6	21:05:05	alter table appointments add check (appointmentDateTime > '2026-04-24 00:00:00')	0 row(s) affected Records: 0 Duplica

SQL Queries and Report

In this part, different SQL queries were applied to retrieve and analyze data from the database.
The queries include SELECT queries, aggregate functions, join query, subquery, and view.

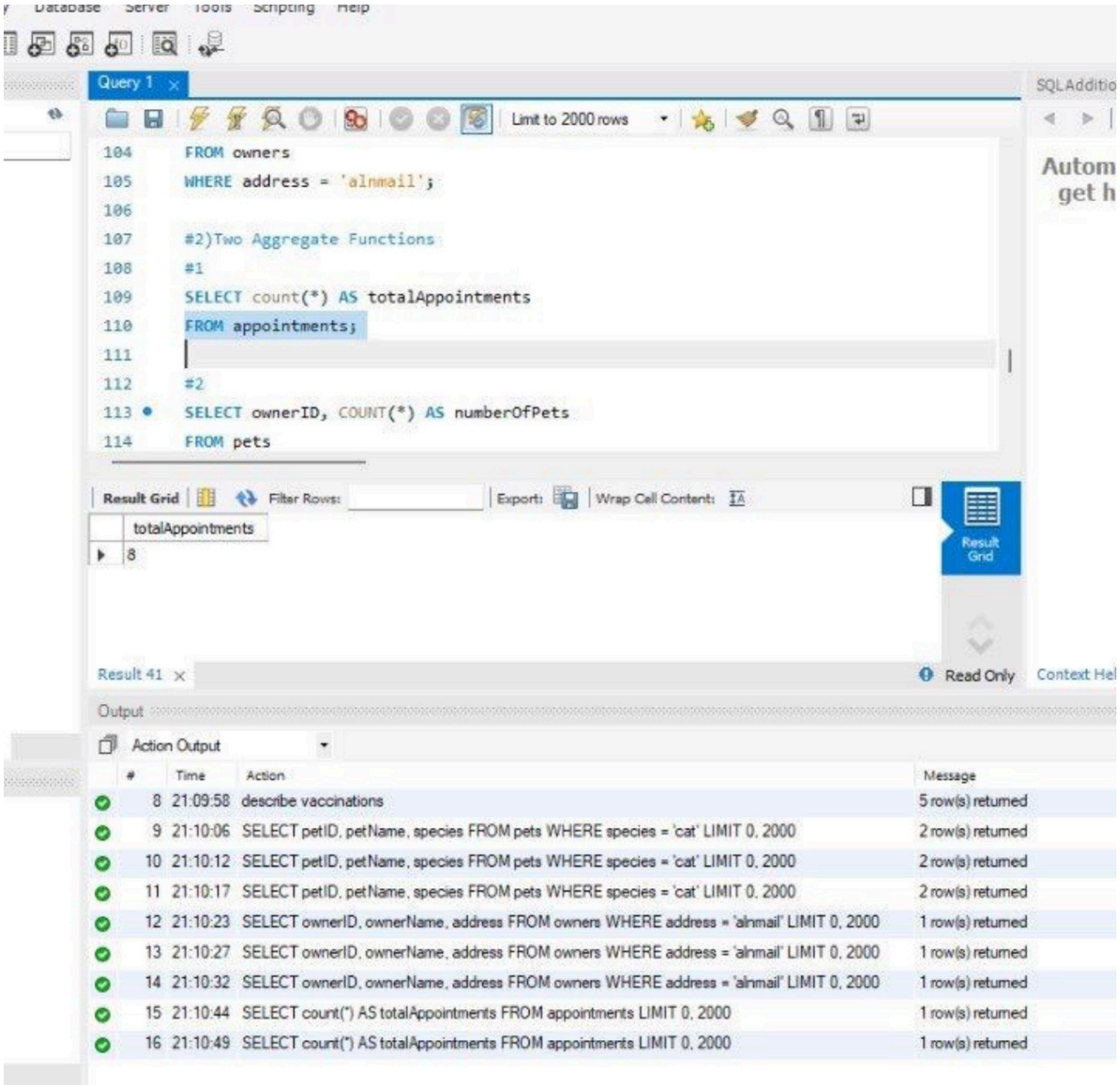
Basic SELECT Query 1

This query displays all pets that have the species cat.



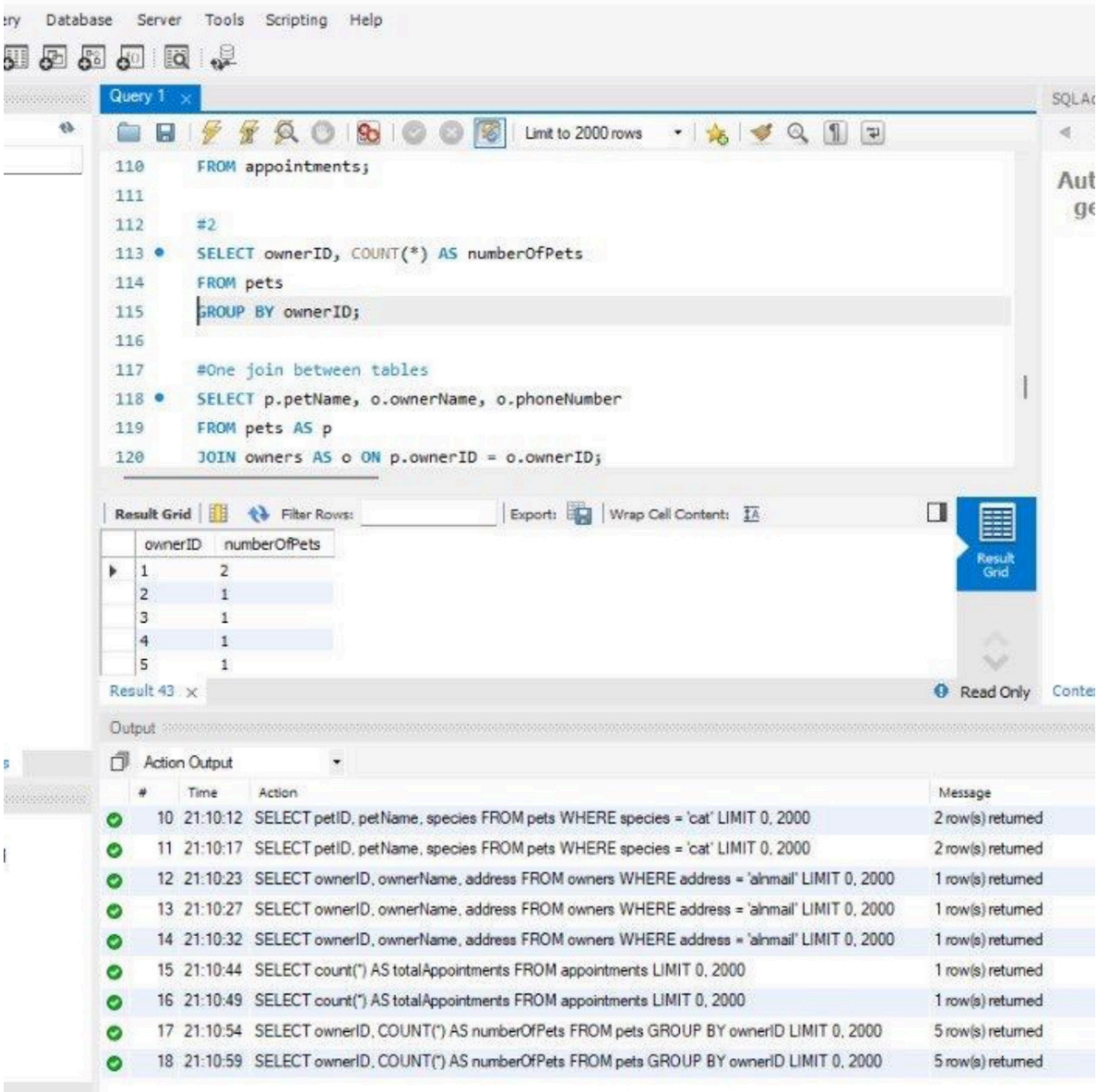
Aggregate Function Query 1

This query counts the total number of appointments in the database.



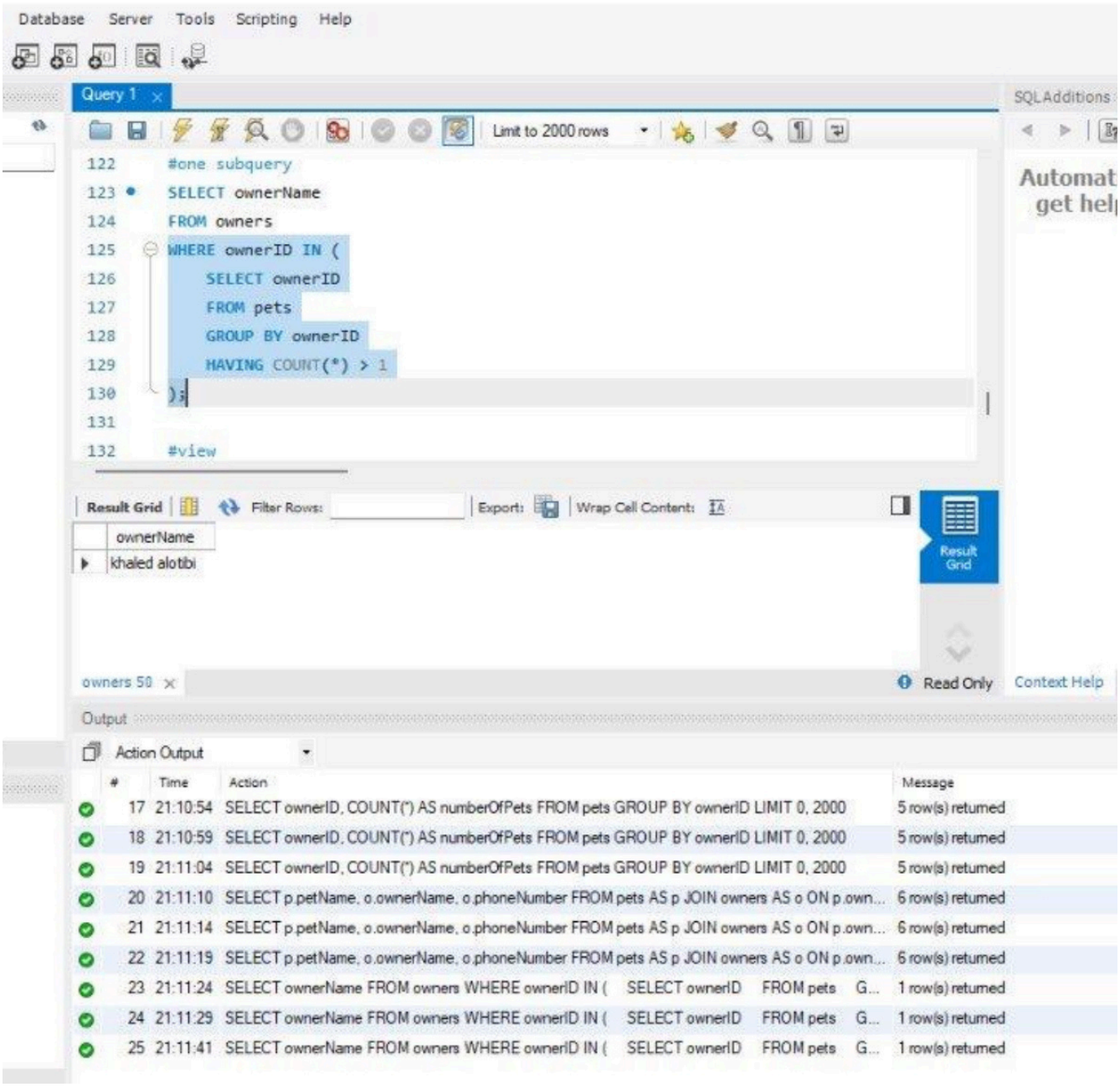
Aggregate Function Query 2

This query counts the number of pets for each owner.



Subquery

This query displays the owners who have more than one pet.





Conclusion

In this project, we designed and implemented a database for a Pet Care Clinic.
We created tables for owners, pets, appointments, and vaccinations.
We also applied different SQL commands such as CREATE, INSERT, SELECT, JOIN, aggregate functions, subquery, and view.
This project helped us understand how to design a database and retrieve useful information from it.